

Ashamoni Kakati

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Summary of Qualifications

Researcher with a Master's in Industrial engineering and 2+ years of experience applying machine learning to solve complex, data driven problems. Skilled in building and deploying Machine Learning models for predictive analytics, risk assessment, and anomaly detection. Proficient in Python and statistical modeling with expertise in end-to-end pipelines from largescale data collection and feature engineering to model deployment and business integration. Adept at collaborating across engineering, product, and quality teams to translate technical insights into measurable business impact. Recognized for intellectual curiosity, mentoring, and delivering high quality solutions that drive decision making efficiency and reduce risk.

Technical Skills

Programming & ML: Python, MATLAB, TensorFlow/Keras, scikitlearn, Pandas, NumPy

Machine Learning & AI: CNN, LSTM, ANN, fuzzy logic systems, predictive modeling, anomaly detection

Data Science & Analytics: Feature engineering, data preprocessing, timeseries forecasting, statistical modeling, probability theory

Tools & Platforms: Jupyter, Power BI, Excel dashboards, Git, SQL basics

Other: Lean Six Sigma, DOE/Taguchi methods, SPC, statistical analysis

Education

- Caterpillar College of Engineering, Bradley University

Peoria, IL, USA

Master of Science in Industrial Engineering; GPA: 4.00/4.00

Aug 2023- May 2025

National Honors: Phi Kappa Phi , Alpha Phi Mu

Research Thesis: Automated quality monitoring in CNC Turning Centre with Live Tooling operations; developed an end-to-end pipeline (data collection+ signal processing + machine learning) to forecast surface roughness & perpendicularity in live-tool milling and drilling via accelerometer sensor integration.

- Visvesvaraya Technological University

Karnataka, India

Bachelor of Engineering in Information Science and Engineering

Thesis: Secure and Practical Outsourcing of Linear Programming in Cloud Computing

Relevant Work Experience

Researcher & Teaching Assistant | Jan 2024 – Present

Bradley University – Industrial & Manufacturing Engineering Dept., Peoria, IL

- Predictive modeling:** Developed CNNLSTM models to forecast surface roughness and perpendicularity using timeseries vibration signals; improved prediction accuracy over conventional methods
- Risk assessment algorithms:** Designed fuzzy logic and ANN frameworks for real-time quality control, reducing out-of-spec outcomes and improving decision reliability
- End-to-end pipelines:** Collected and processed largescale timeseries sensor datasets (500k+ rows/signal); implemented feature extraction, data cleaning, and model integration into SOPs/control plans
- Cross functional collaboration:** Worked with faculty and student teams to align ML-driven insights with process improvement and manufacturing outcomes
- Mentorship:** Guided students in machine learning, DOE, and statistical modeling for applied projects; provided hands-on instruction in Python and modeling tools

Quality Control Engineer Intern | May 2024 – Aug 2024

Micro Products & Company, Peoria, IL

- Machine Learning based quality analysis:** Applied DOE and statistical modeling to identify root causes of process variation; integrated predictive checks into inspection workflows
- Data driven improvement:** Leveraged FMEA and statistical analysis to cut rework and align inspection methods with predictive quality models
- Collaboration:** Partnered with design, production, and quality teams to launch APQP/PPAP activities with data backed insights

Virginia Tech, Blacksburg, VA

Assistant Operations Manager | Feb 2022 – Jan 2023

- **Data analytics & automation:** Introduced macros and forms to automate repetitive workflows, reducing errors and improving data reliability
- **Predictive supply insights:** Analyzed vendor performance data to forecast risk of delivery delays and implemented corrective actions
- **Team leadership:** Coordinated internal teams and vendors, applying analytical decision-making to meet KPIs

IT Consultant | Oct 2020 – Jul 2021

Sports Authority of Assam | India

- **Web Application Development** – Worked in a design and development team of *Online Admission for Regular Coaching Membership in Different Sports* web application. Technologies used: Java, Spring Boot, PostgreSQL, NetBeans
- **Data Analysis & Reporting** – Conducted detailed analysis, generated insights and visualizations, and published reports based on organization-wide data
- **Website Functionality Enhancement** – Developed, reviewed, and improved the functionality of the Sports Authority of Assam website
- **System Upgrades & Technology Planning** – Recommended and installed upgrades; assisted the organization in planning for technology to match growth
- **Database Integrity & Issue Resolution** – Researched and resolved issues regarding the integrity of data flow into databases.
- **Database & Visualization Development** – Developed tables, views, and materialized views using SQL; created dashboards and reports using Spreadsheet and Tableau

Journal Publications

- Joseph C. Chen and **Ashamoni Kakati**, “Development of e-Dart based **Artificial Neural Network** for Multiple Quality Characteristic Online Defect Detection System in Injection Molding Processes.” *The International Journal of Advanced Manufacturing Technology*, 136(3): 951–960, 2025
- Joseph C. Chen and **Ashamoni Kakati**, “**Fuzzy Logic** Framework for Multi-Dimensional Quality Control Online Prediction (FL-MQOP) in Injection Molding Process.” *International Journal on Interactive Design and Manufacturing (IJIDeM)* (under review)
- **Ashamoni Kakati** and Joseph C. Chen, “Development of **CNN-LSTM** Framework for Predicting Surface Roughness and Perpendicularity in Milling and Drilling Operation in Live Tool CNC Lathe Using Accelerometer.” *The International Journal of Advanced Manufacturing Technology* (under review)

Projects

Six Sigma Quality Improvement – Injection Molding: Led a DMAIC/Taguchi/SPC/FMEA project that lifted Cp/Cpk (0.96/0.86 → >benchmark) and cut defects to 0.08%; updated SOPs/control plans for sustained compliance

Facility Layout & Line Balancing (Factory Flow/AutoCAD): Modeled and rebalanced a 14-dept plant (0.289 min/pc takt); reduced WIP 30% and increased throughput 25%

Machine Vision QC Automation: Built a 13-layer CNN system (99.98% detect, ~90% false positives) to automate injection-molding defect checks and deliver real-time quality KPIs

DFM & APQP – Instrument Clamp: Co-designed and prototyped a clamp (CAD/3D print) in CREO; ran PFMEA/APQP, control plans, and mock PPAP to ensure launch readiness

Logistics & Supply Chain Analytics: Modeled MM ERP inputs/lead times and flagged a 19 units/day spike; recommended JIT/vendor collaboration to align demand and supply

Certification

- Six Sigma White Belt-The Council for Six Sigma Certification
- Six Sigma (Green Belt level) – Academic Training: Completed Six Sigma training as part of Master’s program, with hands-on project on quality/process improvement (2024)
- Department Data Analytics – Virginia Tech (2022)